

Take Control Of Software Management on Windows

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How C4B Benefits Organizations



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- Customer Solutions Manager / Director of Marketing / Idea Guy
- Microsoft MVP - PowerShell
- 20+ years of ~~breaking~~ fixing things



Agenda

- The Problem in Managing Software
- What Makes Chocolatey Different
- How Chocolatey for Business (C4B) Benefits Organizations
- Questions



The Problem



What's the Problem in Software Management?

- Manual Installs
- Environment Inconsistency
- Patching is time-consuming. Bigger environments require more time
- Visibility Limited
- Testing Is Hard



Chaos Ensues

- Broken Windows (see what I did there?) and
- Snowflake machines
 - Systems no one wants to touch for fear of breaking
- Failed Updates
- 2 AM phone calls to fix breakages
- Tired Admins keeping the coffee industry profitable



But...we use tools to help out already



Existing Tools?

Existing tools are great, but typically have gaps in managing software

- Intune / SCCM
- Config Mgmt - Ansible, Chef, Puppet
- Home-grown



Chocolatey Is Different



Differences

- Actual Package Management
- Manage All Software - Not Just Installers
- Automation First
- PowerShell for the Masses
- Integrates with Everything



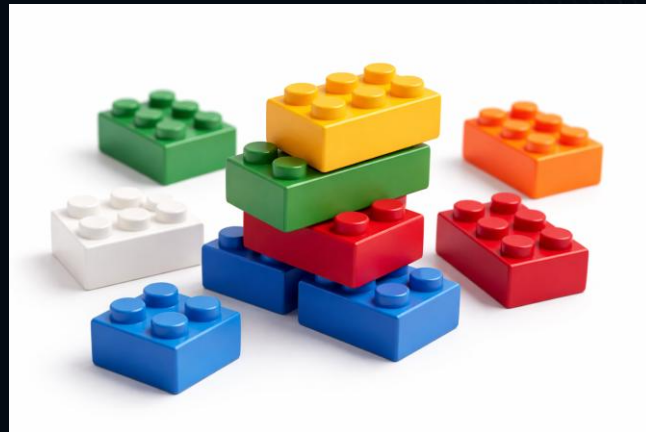
Actual Package Management

- Package != Software
- Package != Installer
- Most other platforms use installers as "packages", but what about tools, LOB apps/websites, files, configuration, etc?
 - MSIs and Installers require expertise to make, don't handle automation well
- Concept of Packaging - Entirely Unique for Windows
 - Like a zip file but
 - Versioning
 - Dependencies
 - Framework for automation
 - PowerShell for the masses



Automation First

- Packages are self-contained deployment blocks
- Test your deployments
- Build consistency & reliability



Chocolatey Packaging

AKA Self-contained deployment

AKA immutable deployment artifact

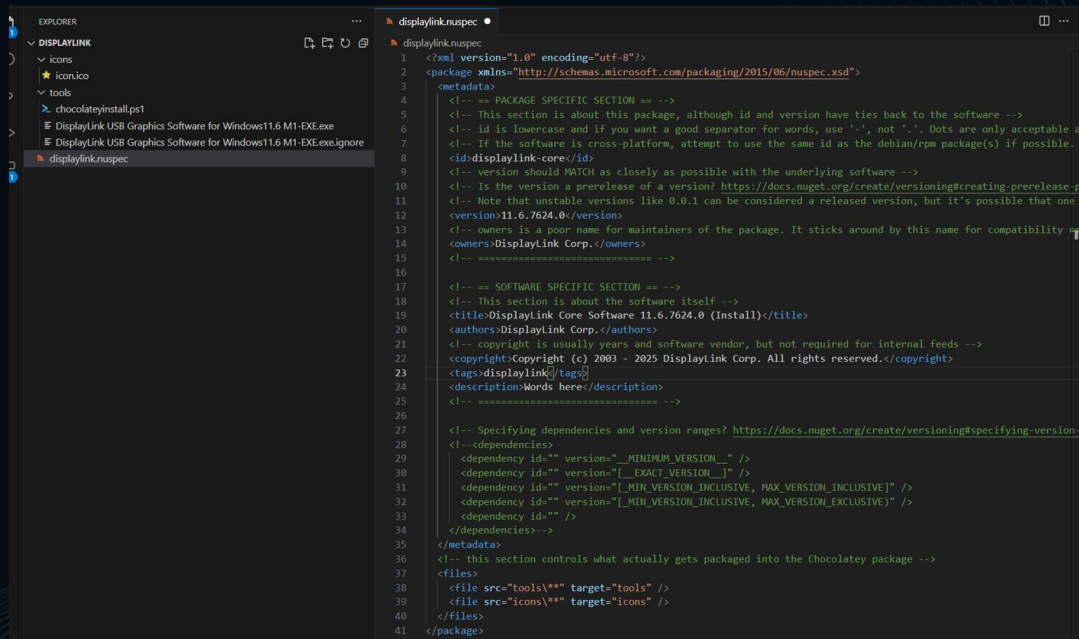
- Metadata - information about the package
- Software and/or files
- Instructions - install / upgrade / uninstall
- Customizations
 - parameters
 - Additional customizations installers may not have
- Defined dependencies on other packages

"A Chocolatey package isn't just files and a script—it's metadata, dependencies, security, and automation wrapped into a reusable unit of infrastructure."



Chocolatey Packaging DNA

- Metadata file (nuspec)
- File/folder contents
 - Installers, tools, files, and/or whatever you want
- Compiled into a "package"



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <package xmlns="http://schemas.microsoft.com/packaging/2015/06/nuspec.xsd">
3   <metadata>
4     <!-- == PACKAGE SPECIFIC SECTION == -->
5     <!-- This section is about this package, although id and version have ties back to the software -->
6     <!-- id is lowercase and if you want a good separator for words, use '-', not '.'. Dots are only acceptable a
7     <!-- If the software is cross-platform, attempt to use the same id as the debian/rpm package(s) if possible.
8     <id>displaylink-core</id>
9     <!-- version should MATCH as closely as possible with the underlying software -->
10    <!-- Is the version a prerelease of a version? https://docs.nuget.org/create/versioning#creating-prerelease-p
11    <!-- Note that unstable versions like 0.0.1 can be considered a released version, but it's possible that one
12    <version>11.6.7624.0</version>
13    <!-- owners is a poor name for maintainers of the package. It sticks around by this name for compatibility ==
14    <owners>DisplayLink Corp.</owners>
15    <!-- ===== -->
16
17    <!-- == SOFTWARE SPECIFIC SECTION == -->
18    <!-- This section is about the software itself -->
19    <title>DisplayLink Core Software 11.6.7624.0 (Install)</title>
20    <authors>DisplayLink Corp.</authors>
21    <!-- copyright is usually years and software vendor, but not required for internal feeds -->
22    <copyright>Copyright (c) 2003 - 2025 DisplayLink Corp. All rights reserved.</copyright>
23    <tags>displaylink</tags>
24    <description>Words here</description>
25    <!-- ===== -->
26
27    <!-- Specifying dependencies and version ranges? https://docs.nuget.org/create/versioning#specifying-version-
28    <!--><dependencies>
29      <dependency id="" version="" _MINIMUM_VERSION_" />
30      <dependency id="" version="[ _EXACT_VERSION_]" />
31      <dependency id="" version="[ _MIN_VERSION_INCLUSIVE, MAX_VERSION_INCLUSIVE]" />
32      <dependency id="" version="[ _MIN_VERSION_INCLUSIVE, MAX_VERSION_EXCLUSIVE]" />
33      <dependency id="" />
34    </dependencies></-->
35  </metadata>
36  <!-- this section controls what actually gets packaged into the Chocolatey package -->
37  <files>
38    <file src="tools\*" target="tools" />
39    <file src="icons\*" target="icons" />
40  </files>
41 </package>
```



Chocolatey Packaging DNA <cont>

- Instructions (PowerShell)

```
s > > chocolateyinstall.ps1 > ...  
$ErrorActionPreference = 'Stop';  
$toolsDir      = "$(Split-Path -parent $MyInvocation.MyCommand.Definition)"  
$fileLocation = Join-Path $toolsDir 'DynUpdater.msi'  
  
$packageArgs = @{  
    packageName     = $env:ChocolateyPackageName  
    softwareName    = 'Dyn Updater*'  
    file            = $fileLocation  
    fileType        = 'msi'  
    silentArgs      = "/qn /norestart /l*v `"$env:TEMP\$env:ChocolateyPackageName.$env:ChocolateyPackageVersion.log`""  
    validExitCodes = @(0,1641,3010)  
    checksum        = '0CCC51A81A422B7925CE8FD18FC6206799D9AF77364BAAE6010102DE0E6BE495'  
    checksumType    = 'sha256'  
    destination     = $toolsDir  
}  
  
Install-ChocolateyInstallPackage @packageArgs
```

- <https://docs.chocolatey.org/en-us/create/functions/install-chocolateyinstallpackage/>



PowerShell for the Masses

- Scripting language of Windows
- PowerShell == Automation
- Shared Platform for Admins and Devs



Integrates with Everything

- Lift and Shift Not Required
 - Deployment Instructions move to packaging
 - Existing platforms shift to simple choco commands
- Allows platforms to manage more than installers
- Same Deployment for Multiple Platforms
 - Ansible to Servers
 - Intune to Desktops
 - Both can use the same Chocolatey packages



What Happens If We Build a Platform to Meet Organizational Needs on Top of Chocolatey?



Organizational Needs for Software Management

- Security, Consistency, & Reliability
- Controlled Inputs
- Manage Many Machines (Endpoint Management)
- Reporting & Auditing
- Provide Controlled Software Autonomy for End Users (Self-Service)
- Ease of Use



Chocolatey for Business (C4B)

- Chocolatey Central Management - Secure, Worldwide Endpoint Management
- Self-Service Anywhere – Control Available Software for Install By End Users
- Package Preparation
 - Package Builder
 - Package Internalizer
- Package Audit - Comprehensive Auditing of Packages
- Package Reducer - Optimized Environments
- Programs & Features Integrations



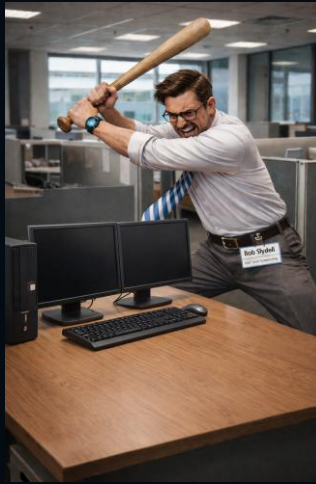
C4B: Chocolatey Central Management - Endpoint Management

- Secure, worldwide deployments at scale
- Centralized endpoint management
- Real-time reporting and visibility
- Improve compliance and governance

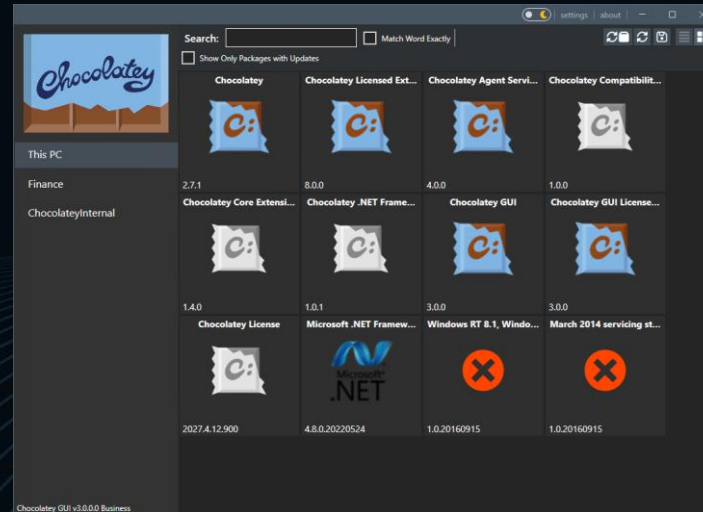


C4B: Self-Service Anywhere

- End Users Install Only Approved Software
- No Admin Privileges Required
- Simple UI
- Frees IT for high-value work
- Works with
 - On site teams
 - Remote users
 - Hybrid work environments



SSA
turns
that
frown
down



C4B: Package Builder

- Generate packages from Installers or Programs and Features
- Reduces manual packaging
- Create packages in seconds

C4B: Package Internalizer

- Pulls existing community packages and converts for offline use
- Reduces supply chain risk
- Reliable, repeatable installs



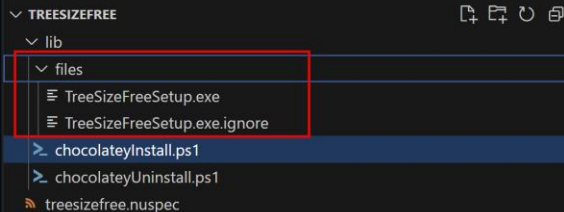


Administrator: Windows Powe



```
PS C:\packages> choco download treesizefree --internalize --source https://community.chocolatey.org/api/v2  
Chocolatey v2.7.1 Business
```





```
lib > > chocolateyInstall.ps1 > > $LicensedCommandsRegistered
1 $LicensedCommandsRegistered = Get-Command "Invoke-ValidateChocolateyLicense" -EA SilentlyContinue
2 if (!$LicensedCommandsRegistered) {
3     Write-Warning "Package Requires Commercial License - Installation cannot continue as Package
4     throw "This package requires a commercial edition of Chocolatey as it was built/internalized
5 }
6
7 Invoke-ValidateChocolateyLicense -RequiredLicenseTypes @('Business','ManagedServiceProvider')
8
9 if ($env:chocolateyPackageParameters -match '/lang=') {$param=$env:chocolateyPackageParameters}
10
11 $packageArgs = @(
12     packageName = 'treesizefree'
13     fileType = 'exe'
14     silentArgs = "$param -verysilent /AllUsers"
15     url = "$(Split-Path -parent $MyInvocation.MyCommand.Definition)\files\TreeSizeFreeSetup.exe"
16     checksum = 'D800F1765539194A088CD0682B8A73C40C4A1001425F2A892F495A53DB9DA2DE'
17     checksumType = 'sha256'
18 )
19
20 $osInfo = Get-CimInstance -ClassName Win32_OperatingSystem
21 if ($osInfo.ProductType -gt 1) {
22     throw "TreeSize Free is not supported on Windows Server systems. Please refer to TreeSize Professional"
```

Hide lib\chocolateyInstall.ps1

```
1 if ($env:chocolateyPackageParameters -match '/lang=') {$param=$env:chocolateyPackageParameters} Else {$pa
2
3 $packageArgs = @(
4     packageName = 'treesizefree'
5     fileType = 'exe'
6     silentArgs = "$param -verysilent /AllUsers"
7     url = 'https://downloads.jam-software.de/treesize_free/archive/v4.8.1.610/TreeSizeFreeSetup.exe'
8     checksum = 'D800F1765539194A088CD0682B8A73C40C4A1001425F2A892F495A53DB9DA2DE'
9     checksumType = 'sha256'
10 }
11
12 $osInfo = Get-CimInstance -ClassName Win32_OperatingSystem
13 if ($osInfo.ProductType -gt 1) {
14     throw "TreeSize Free is not supported on Windows Server systems. Please refer to TreeSize Professional
15 }
16
17 Install-ChocolateyPackage @packageArgs
```

```
$packageArgs["UseOriginalLocation"] = $true
Install-ChocolateyPackageCmdlet @packageArgs
```

Chocolatey

C4B: Package Audit

- Comprehensive Auditing of Packages
- What, version in Open Source
- Who, when, where

```
-----
13 packages installed.
PS C:\> choco list --audit -r
chocolatey|2.7.1||||
chocolatey.extension|8.0.0|||||https://chocolatey.willywonka.dev:8443/repository/ChocolateyInternal/index.json
chocolatey-agent|4.0.0|admin|CCMVENDORCLI|admin|2026-04-11 15:46:40Z|https://chocolatey.willywonka.dev:8443/repositor
y/ChocolateyInternal/index.json
chocolatey-compatibility.extension|1.0.0|admin|CCMVENDORCLI|admin|2026-04-11 15:44:24Z|https://chocolatey.willywonka.
dev:8443/repository/ChocolateyInternal/index.json
chocolatey-core.extension|1.4.0|admin|CCMVENDORCLI|admin|2026-04-11 15:44:26Z|https://chocolatey.willywonka.dev:8443/
repository/ChocolateyInternal/index.json
chocolatey-dotnetfx.extension|1.0.1|admin|CCMVENDORCLI|admin|2026-04-11 15:44:28Z|https://chocolatey.willywonka.dev:8
443/repository/ChocolateyInternal/index.json
chocolateygui|3.0.0|admin|CCMVENDORCLI|admin|2026-04-11 15:45:48Z|https://chocolatey.willywonka.dev:8443/repository/C
hocolateyInternal/index.json
chocolateygui.extension|3.0.0|admin|CCMVENDORCLI|admin|2026-04-11 15:46:02Z|https://chocolatey.willywonka.dev:8443/re
pository/ChocolateyInternal/index.json
chocolatey-license|2027.4.12.900|||||https://chocolatey.willywonka.dev:8443/repository/ChocolateyInternal/index.json
dotnetfx|4.8.0.20220524|admin|CCMVENDORCLI|admin|2026-04-11 15:45:27Z|https://chocolatey.willywonka.dev:8443/reposito
ry/ChocolateyInternal/index.json
KB2919355|1.0.20160915|admin|CCMVENDORCLI|admin|2026-04-11 15:44:38Z|https://chocolatey.willywonka.dev:8443/repositor
y/ChocolateyInternal/index.json
KB2919442|1.0.20160915|admin|CCMVENDORCLI|admin|2026-04-11 15:44:37Z|https://chocolatey.willywonka.dev:8443/repositor
y/ChocolateyInternal/index.json
microsoft-onedrive|26.51.316.4|admin|CCMVENDORCLI|sync|2026-04-13 20:00:21Z|
PS C:\>
```



C4B: Programs & Features Synchronization

- Choco Sync
- All Current Installed Software – Now Managed by Chocolatey
- Take Over in Seconds

```
PS C:\> choco list -i
Chocolatey v2.7.1 Business
5 validations performed. 4 success(es), 1 warning(s), and 0 error(s).

chocolatey 2.7.1
chocolatey.extension 8.0.0
chocolatey-agent 4.0.0
chocolatey-compatibility.extension 1.0.0
chocolatey-core.extension 1.4.0
chocolatey-dotnetfx.extension 1.0.1
chocolateygui 3.0.0
chocolateygui.extension 3.0.0
chocolatey-license 2027.4.12.900
dotnetfx 4.8.0.20220524
KB2919355 1.0.20160915
KB2919442 1.0.20160915
microsoft-onedrive 26.51.316.4
13 packages installed.

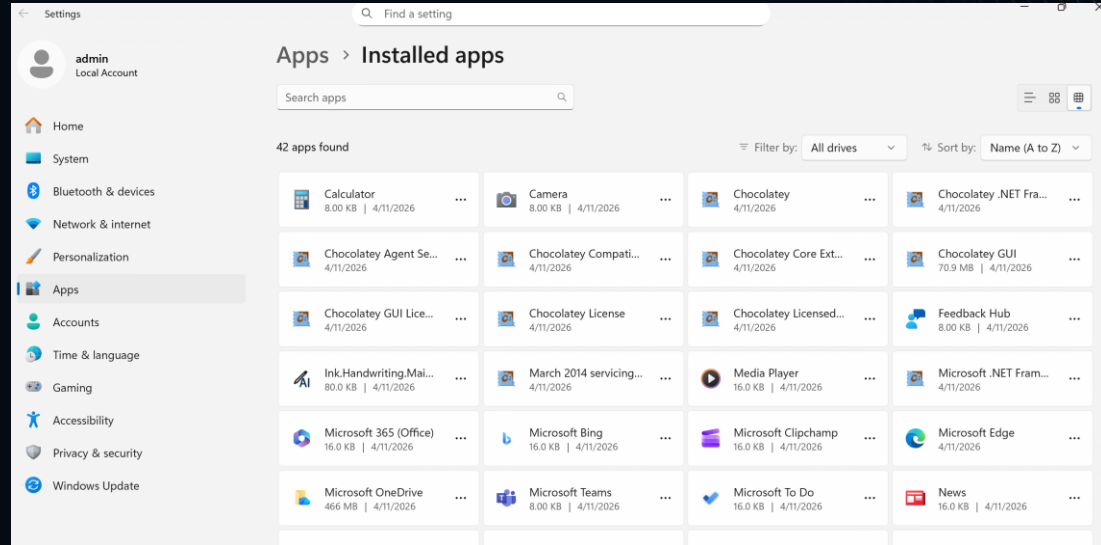
0 applications not managed with Chocolatey.
PS C:\>
```

1



C4B: Programs & Features Synchronization

- All packages show in Programs and Features
- Additionally shows everything being managed with Chocolatey
 - Not Just Installers
- Enhances Security and Reporting



Demo



Questions?

- <https://chocolatey.org>
- Talk to Us - <https://chocolatey.org/contact>
- Come By the Booth! (We have “stuff”)

